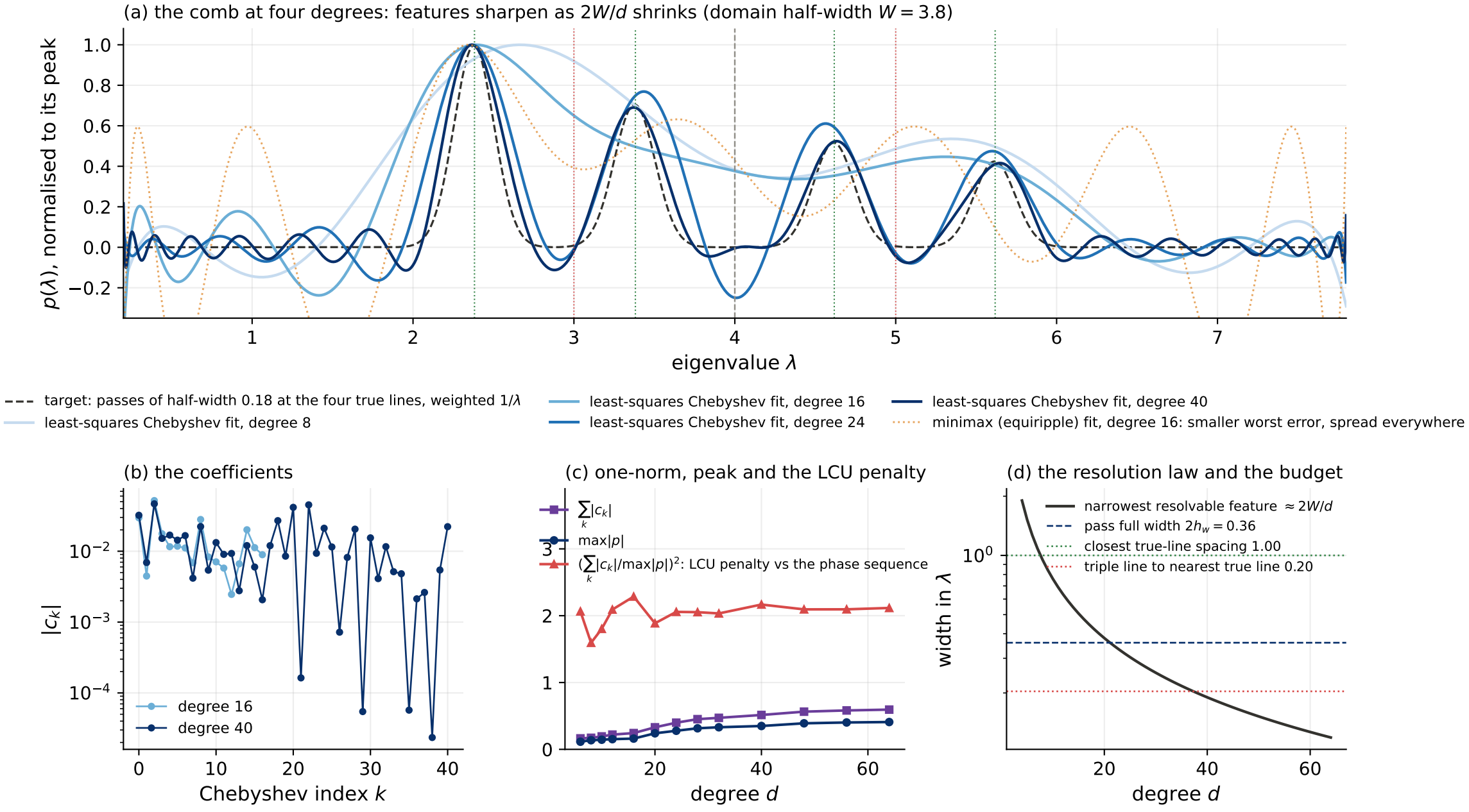
(a) the comb at four degrees: features sharpen as 2W/d shrinks (domain half-width W = 3.8)
p(λ), normalised to its peak
eigenvalue λ
target: passes of half-width 0.18 at the four true lines, weighted 1/λ
least-squares Chebyshev fit, degree 8
least-squares Chebyshev fit, degree 16
least-squares Chebyshev fit, degree 24
least-squares Chebyshev fit, degree 40
minimax (equiripple) fit, degree 16: smaller worst error, spread everywhere
(b) the coefficients
|c_k|
degree 16
degree 40
Chebyshev index k
(c) one-norm, peak and the LCU penalty
Σ_k |c_k|
max|p|
(Σ_k |c_k|/max|p|)²: LCU penalty vs the phase sequence
degree d
(d) the resolution law and the budget
narrowest resolvable feature ≈ 2W/d
pass full width 2h_w = 0.36
closest true-line spacing 1.00
triple line to nearest true line 0.20
width in λ
degree d